

## Handout 02.12: Compound Growth Table

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Partner: \_\_\_\_\_

The rate on this sheet is 5 percent a year so the math is clean. Real savings accounts in 2026 pay from under 1 percent at big banks to about 4 percent at online banks.

**My guess:** \$20 a week for 30 years at 5 percent = \$\_\_\_\_\_

### Part A: Why save, and where

Three reasons to save, with an example of each:

| Reason               | My example |
|----------------------|------------|
| 1. A goal            |            |
| 2. An emergency fund |            |
| 3. The future        |            |

Jar versus savings account:

|                                                             | A jar or a drawer | A savings account |
|-------------------------------------------------------------|-------------------|-------------------|
| Does it grow?                                               |                   |                   |
| Is it protected if it is lost or stolen, or the bank fails? |                   |                   |
| How easy is it to raid?                                     |                   |                   |
| Who can open one?                                           |                   |                   |

**Pay yourself first** means \_\_\_\_\_

### Part B: Simple versus compound (copy from the slide)

\$1,000 at 5 percent.

Simple interest: \$50 a year, every year. After 3 years: \$1,000 + \$150 = \$\_\_\_\_\_

Compound interest:

Year 1: \$1,000 x 0.05 = \$\_\_\_\_\_ interest. End: \$\_\_\_\_\_

Year 2: \$\_\_\_\_\_ x 0.05 = \$\_\_\_\_\_ interest. End: \$\_\_\_\_\_

Year 3: \$\_\_\_\_\_ x 0.05 = \$\_\_\_\_\_ interest. End: \$\_\_\_\_\_



The shortcut:  $\$1,000 \times 1.05 \times 1.05 \times 1.05 = \$1,000 \times 1.05$  to the third power = \$\_\_\_\_\_

### Part C: \$20 a week, by hand

\$20 a week  $\times$  52 weeks = \$\_\_\_\_\_ a year. You deposit it at the start of each year. At the end of the year the bank pays 5 percent on everything in the account.

| Year | In the account at the start | Add this year | Balance before interest | The bank pays (5 percent)          | End of year |
|------|-----------------------------|---------------|-------------------------|------------------------------------|-------------|
| 1    | \$0.00                      | \$1,040.00    | \$1,040.00              | $\$1,040.00 \times 0.05 = \$52.00$ | \$1,092.00  |
| 2    | \$1,092.00                  | \$1,040.00    | \$_____                 | $\$_____ \times 0.05 = \$_____$    | \$_____     |
| 3    | \$_____                     | \$1,040.00    | \$_____                 | $\$_____ \times 0.05 = \$_____$    | \$_____     |
| 4    | \$_____                     | \$1,040.00    | \$_____                 | $\$_____ \times 0.05 = \$_____$    | \$_____     |
| 5    | \$_____                     | \$1,040.00    | \$_____                 | $\$_____ \times 0.05 = \$_____$    | \$_____     |

The jar after 5 years:  $\$1,040 \times 5 = \$_____$

The account after 5 years: \$\_\_\_\_\_

The difference (the bank paid you this much for waiting): \$\_\_\_\_\_

Why does year 5 earn more interest than year 2 when the deposit is the same?

(Grade 6: years 1 to 3, and the jar at 3 years: \$3,120.)

**Part D: The reveal (copy from the slide)**

| Years | The jar  | The account at 5 percent | The difference |
|-------|----------|--------------------------|----------------|
| 5     | \$5,200  | \$                       | \$             |
| 10    | \$10,400 | \$                       | \$             |
| 30    | \$31,200 | \$                       | \$             |

My guess was \$\_\_\_\_\_. The real number was \$\_\_\_\_\_.

**Two savers.** Nia puts \$1,000 in an account at 15 and never adds to it. At 5 percent she has about \$11,467 at 65. Cole puts \$2,000 in at 35. At 5 percent he has about \$8,644 at 65.

Who ended with more? \_\_\_\_ Who put in more? \_\_\_\_ What mattered more, the amount or the years? \_\_\_\_

### Part E: My pay-yourself-first plan (three sentences)

1. I will save \$\_\_\_\_\_ a week (or \$\_\_\_\_\_ a month).
2. I will keep it in \_\_\_\_\_ (a jar, a youth savings account with an adult, another place) because \_\_\_\_\_.
3. My first goal for it is \_\_\_\_\_ by \_\_\_\_\_.

### Grade 8 stretch

1. Redo the five-year table at 3 percent on the back. End of year 5: \$\_\_\_\_\_. How much does the rate matter over five years? \$\_\_\_\_\_ less.
2. Check Nia's number with the shortcut:  $\$1,000 \times 1.05$  to the 50th power. (Use the calculator's power key.) \$\_\_\_\_\_
3. Three places for the money: a big bank savings account at 0.5 percent, an online bank at 4 percent, a credit union at 3 percent. Which pays the most? \_\_\_\_\_ Name two things besides the rate that should decide where the money goes: \_\_\_\_\_, \_\_\_\_\_

### Exit card (3-2-1)

3 reasons to save: \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_

2 differences between a jar and a savings account: \_\_\_\_\_, \_\_\_\_\_

1 number: \$20 a week for 30 years at 5 percent: \$\_\_\_\_\_